

DJ Desk Decisions

Architecture records and core product notes.

docs/adr/0001-application-backbone.md

ADR 0001: Application Backbone

Status

Accepted.

Date

2026-04-20

Context

DJ Desk needs to move quickly from raw track analysis into an interactive set-planning workspace. The first product risk is the domain model: keys, modes, BPM, chord progressions, confidence states, and set planning workflows. Rendering performance is important, but it is not the bottleneck yet.

The application starts local-first, but it should be deployable later without changing the core architecture.

Decision

Use an npm workspace monorepo:

- apps/client: Vue 3 + Vite application.
- apps/server: Fastify API running TypeScript directly on Node 24 LTS.
- packages/domain: shared TypeScript package for musical domain types and helpers.
- docs/adr: architecture decisions.

Use Node 24 LTS as the runtime baseline and npm as the only package manager. Do not use Yarn Plug'n'Play or pnpm.

Use Vue 3 stable first. Do not base the application on Vue Vapor Mode yet, because Vapor is still beta-only in Vue 3.6. Keep the client in Composition API and `<script setup>` so Vapor can be evaluated later for isolated, performance-sensitive surfaces.

Use TypeScript 6 as the stable compiler. Keep TypeScript 7 native preview optional through `npm run typecheck:native`; it is not a required build step.

Use Oxlint and Oxfmt as the primary linting and formatting tools. Avoid ESLint/Prettier until a concrete rule gap appears.

Use a build boundary for packages/domain. Node can run application .ts files directly, but built-in type stripping refuses TypeScript files under node_modules. The shared package therefore emits JavaScript and declarations to dist.

Prepare the backend for SQLite persistence, but keep the initial API in memory while the domain model and screens are still forming.

Consequences

- The development stack stays small and fast.
- The server can run `.ts` directly with Node's stable type stripping.
- Type checking remains explicit via `tsc`; Node does not type-check runtime TypeScript.
- Shared domain logic is isolated from framework code and can be tested independently.
- The app remains easy to deploy later: build the domain package, build the client, and run the Fastify server on Node 24 LTS.
- Vapor Mode can be tested later without forcing the whole client onto a beta runtime.

ADR 0002: Modal Circle Workspace

Status

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Date

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Context

The main DJ Desk surface is a production workspace for manually arranging tracks into a set. The interface must help the operator reason about harmonic proximity quickly, while still leaving room for manual judgement.

Most tracks can be placed by tonic and common mode, but Brazilian and adjacent repertoire often contains modal color, variable scale degrees, mixture, or unusual modulations.

Those cases need to be visible without forcing a full music-theory analysis into the main map.

The future dataset is expected to contain hundreds of tracks, so the first screen must support dense browsing, quick focus changes, and a draft set chain.

Decision

Use a twelve-section circle-of-fifths workspace as the primary visual model. The first solmization labels are fixed to Do, Re, Mi, Fa, Sol, La, Si, with enharmonic spellings shown together for sharp/flat pitch classes.

Track placement is derived from `TrackKey` and represented as `ModalPlacement`:

- `home`: major and natural minor tracks remain in the tonic section.
- `pure-modal`: pure Dorian, Lydian, Phrygian, and Mixolydian tracks are placed in the section representing their diatonic collection.
- `modal-mixture`: tracks with a variable degree are placed between the home tonic and the pure modal target section.

Use `variable-degree` as the data variant and `modal mixture` as the UI lane language. This avoids calling tracks "dirty" while still capturing the practical mixing concern.

Add two independent free-text fields to tracks:

- `harmonyNotes`: special warnings or instructions about harmony, modulation, exotic modes, or other analysis details that require attention before mixing.
- `comment`: ordinary operator notes that do not need warning treatment.

When `harmonyNotes` is present, show a compact warning indicator on the track dot, in the sector browser, and inside the draft set chain. Show the full text only in the focused track panel.

Keep chord progressions as a simple list for now. The future raw dataset may contain a large amount of chord data, so the detailed chord table will be designed after import format and density are known.

Consequences

- The map stays fast to scan even when tracks have nuanced modal behavior.
- A track can be visible near both its home tonic and its modal collection through the modal-mixture lane and nearby-section filtering.
- Special harmony warnings are not lost inside ordinary comments.
- The model remains small enough for the spike while leaving room for richer analysis data, chord tables, solmization preferences, and import workflows.

ADR 0003: SQLite Persistence and Migrations

Status

Accepted.

Date

2026-04-20

Context

DJ Desk starts as a local-first application, but it should not keep its core data only in memory. Track analysis, harmony notes, comments, chord lists, tags, and draft set order need a durable structure before raw import work begins.

The app is still a spike, so the persistence layer should stay small and easy to replace if deployment requirements later push the project toward Postgres or another database.

Decision

Use SQLite as the first durable store. Store the default database at `data/djdesk.sqlite` and allow overrides through `DATABASE_PATH`.

Use plain SQL migration files under `apps/server/migrations`. The Fastify server applies pending migrations on startup, and `npm run db:migrate` runs the same migration path manually.

Keep migrations structural only. Seed data comes from the existing TypeScript sample tracks and is inserted only when the database is empty. This keeps production data and local bootstrap data separate.

Create normalized tables for:

- tracks and their key, BPM, confidence, harmony note, and comment metadata.
- ordered chord symbols per track.
- tags per track.
- draft sets and ordered draft set tracks.

Consequences

- Local development has a durable database without adding an external service.
- The API can move from in-memory sample data to real persistence without changing the client contract.
- Raw imports can target normalized tables instead of pushing large JSON blobs through the app.
- Future migration to another SQL database remains possible because schema changes are explicit and repository access is isolated on the server.

ADR 0004: Modal Subsections and Transition Rules

Status

Accepted.

Date

2026-04-20

Context

The imported catalog is dense enough that dots on the modal circle are not a usable primary control. DJ Desk needs larger tap targets and a clearer model for pure modal tracks and modal-mixture tracks around each circle boundary.

The operator also needs the Add warning and wheel coloring to follow a small, explicit transition graph instead of a broad "nearby key" heuristic.

Decision

Keep the twelve natural-minor circle sections as the main frame.

Within each section, divide the annular area into:

- a central home subsection for natural minor, major via relative minor, and raised-leading-tone home tracks.
- two side pure-modal subsections near the neighboring boundaries.
- a boundary subsection between each pair of circle sections for modal-mixture tracks.

Pure modal tracks are displayed in the section of the natural-minor collection they use, not in the section of their tonic. For example:

```
La natural minor
Mi pure Phrygian
La Dorian mixture + Mi Phrygian mixture
La pure Dorian
Mi natural minor
```

Use these directed transition rules for the draft-set Add warning:

- home can move to the same or neighboring home section.
- pure modal can move to pure modal with the same mode and same or neighboring home tonic.
- home can move to pure modal when the home section is the pure modal track's home section or target collection section.
- pure modal can move to home when the home section is the pure modal track's home section or target collection section.
- home can move to modal mixture when the home section belongs to that modal-mixture boundary.
- modal mixture can move to home when the home section belongs to that modal-mixture boundary.
- modal mixture can move to modal mixture only on the same boundary.

- modal mixture can move to pure modal only for the same home tonic and same mode.

The wheel compatibility coloring stays symmetric for browsing: a zone is shown as compatible with the last draft track when at least one track in the zone can move in either direction with that draft endpoint.

Consequences

- Pure modes are easier to understand visually because they sit beside the natural-minor collection they actually use.
- Modal-mixture tracks from neighboring tonics naturally share one boundary target.
- The add button remains directional, while the wheel stays useful for browsing both possible directions around the current draft endpoint.
- The rules are intentionally conservative and can be revised after listening tests without changing the storage model.

ADR 0005: Default Key Display Solmization

Status

Accepted.

Date

2026-04-20

Context

DJ Desk stores analyzed keys as normalized pitch classes so imports, transition rules, and database queries stay compact and stable. That internal representation should not dictate how the interface displays keys.

The working UI should use the operator's default solmization vocabulary instead of raw letter names. Letter-key display may return later as an explicit display preference, but it should not leak into the default interface.

Decision

Keep `TrackKey.tonic` and database `tonic` values in normalized pitch-class form.

Use domain display formatters for user-facing key labels:

- C -> Do
- D -> Re
- E -> Mi
- F -> Fa
- G -> Sol
- A -> La
- B -> Si

For altered pitch classes, show Unicode accidentals:

- sharp: $\sharp$
- flat: $\flat$

When an imported key is stored in the sharp-normalized internal form but has a common enharmonic flat display, show both solmization labels in the UI. The canonical spelling is shown first and the alternate spelling second.

Circle section labels keep the minor-section suffix, for example `La m` and `Fa $\flat$  m` / `Sol $\flat$  m`. Section labels use the natural-minor circle convention: `Do $\sharp$  m` remains primary for C-sharp minor, while `Mi $\sharp$  m` and `Si $\sharp$  m` are primary for the D-sharp/E-flat and A-sharp/B-flat sections.

Full track key labels combine the solmization tonic with the mode label, for example `Re Dorian`. For full key labels, choose the canonical tonic spelling by the key's mode spelling cost: prefer the spelling that avoids excessive accidentals in the full seven-note mode. For example:

- C-sharp natural minor -> `Do $\sharp$` / `Re $\sharp$  Natural minor`

- C-sharp major -> Re■ / Do■ Major
- A-sharp natural minor -> Si■ / La■ Natural minor

Consequences

- The UI no longer exposes raw A, B, C, D, or E key names by default.
- Enharmonic pairs no longer always put the sharp spelling first.
- Importers and transition logic do not need storage changes.
- A future notation switcher can be implemented as another formatter without changing the database schema.

ADR 0006: Dense Set-Planning Interactions

Status

Accepted.

Date

2026-04-20

Context

After importing real catalog data, the modal circle is no longer a small demo map. Each section can contain many tracks, some tracks have unknown keys, and a draft set can become harmonically invalid after removing or moving one item.

The operator needs the interface to make these states visible without blocking manual judgement. DJ Desk should warn about risky transitions, but it should not prevent intentional non-harmonic choices.

The first implementation remains local and client-side. Draft set editing is mocked in the Vue state until persisted multi-set editing is implemented.

Decision

Keep track browsing centered on sections, subsections, and boundaries, but make the selected map area explicit:

- selected parent sections, natural subsections, pure-modal subsections, and boundary subsections receive a visible selected contour.
- the center readout describes the selected map area, not the currently focused track.
- boundary readouts use compact primary section labels so long enharmonic strings do not overflow the center of the circle.

Treat unknown-key tracks as a separate shelf outside the circle:

- the unknown-key control is placed at the bottom-right of the wheel panel.
- unknown-key tracks remain hidden from harmonic sections until a key is known.
- selecting the shelf opens only unplaced tracks in the browser.

Use user-facing music language in the browser:

- show `natural` tracks instead of the internal `home` tracks label.
- show `natural`, `pure modal`, `modal mixture`, and `unknown key` as lane labels on track cards.

Keep the Add action permissive but stable:

- a red Add button means the selected track is already present in any draft set, the transition is non-harmonic, or both.
- the visible reason is shown next to the button.
- the action area reserves a fixed layout slot so the button does not jump when the reason appears or disappears.

- repeated tracks are allowed because a DJ set may intentionally reuse a track.

Expose draft set editing directly in the chain:

- tracks can be removed from any position.
- tracks can be moved with up/down controls.
- tracks can also be reordered with native drag-and-drop.
- after any edit, every adjacent transition is re-evaluated.
- risky adjacent pairs are shown as red Non-harmonic transition dividers between the two tracks, because non-harmonicity is a relationship between neighboring set positions rather than an attribute of either track alone.

Add a Sort harmonically control for the active draft set:

- the sorter is a deterministic client-side helper, not a persisted planner.
- it tries each track as a possible start and greedily chooses the next best track by the current transition rules.
- compatible transitions score highest; BPM distance and circle distance are only tie-breakers.
- if no fully compatible sequence exists, the best available order is applied and remaining risky joins stay visibly marked.

Add BPM compatibility to the track browser:

- browser tracks are sorted against the last track in the active draft set.
- the primary browser sort is a soft BPM compatibility score, followed by harmonic compatibility as a tie-breaker.
- the BPM score compares the real BPM values directly; half-time and double-time equivalence are intentionally not considered because this workflow does not mix that way.
- the score is displayed as a number and clipped progress meter on each track card.
- the progress meter uses one fixed red-yellow-green scale and clips the unfilled right side, so short bars do not rescale the full gradient.

Keep verification confidence close to the values it qualifies:

- track cards do not show large standalone Key unverified or BPM unverified badges.
- key and BPM confidence labels are rendered inline next to the corresponding key or BPM value.
- the focus panel uses the same inline confidence pattern.

Keep the future focused-view idea documented but out of scope for this change:

a later map mode may show only the active sector and neighboring sectors instead of the full circle, improving tap targets on small screens.

Consequences

- Dense sections are easier to operate because the selected browsing target is visible independently from the last draft track.
- Unknown-key tracks are no longer presented as if they belonged to the harmonic map.
- The operator can intentionally create, inspect, and repair non-harmonic joins.

- Non-harmonic warnings are easier to interpret because they are attached to transitions, not track cards.
- BPM proximity becomes visible during next-track browsing without imposing a hard compatibility cutoff.
- Track cards are less visually noisy while still preserving unverified key and BPM information.
- The draft set chain becomes editable without introducing a persistence layer change yet.
- Harmonic sorting is useful as a fast repair tool but remains explainable and non-authoritative.
- Future work should persist draft set changes, add undo/redo, and consider a focused sector view for mobile and laptop layouts.

ADR 0007: Local Audio Playback

Status

Accepted.

Date

2026-04-20

Context

DJ Desk now stores the real track catalog, and the operator needs to listen to tracks while planning a set. The audio files already exist on the local machine and should remain the source of truth. The app should not copy or re-encode them into the project.

Browsers cannot reliably play arbitrary local filesystem paths from a web app, and direct `file://` URLs would not be suitable once the app moves from local hosting to a server.

Decision

Store an optional absolute `audio_path` on tracks.

Add a repeatable audio-linking CLI:

- root command: `npm run audio:link -- --root <music folder>`.
- server command: `audio:link`.
- the linker recursively scans configured roots for audio files.
- supported extensions are `.aac`, `.aif`, `.aiff`, `.flac`, `.m4a`, `.mp3`, and `.wav`.
- files are matched against imported ChordAI filenames, track titles, and artist-title combinations.
- matching normalizes punctuation, accents, underscores, dashes, safe join-word differences, and optional filename extensions.
- the database stores the chosen absolute path and does not copy the file.

Expose playback through Fastify, not through direct filesystem URLs:

- `GET /api/tracks/:trackId/audio` streams the linked file.
- the route only accepts a track id and reads the path from SQLite.
- range requests are supported so browser controls can seek within long tracks.
- content type is inferred from the linked file extension.

Keep the first UI implementation simple:

- the track browser shows a compact audio chip when a source file is linked.
- the focus panel shows a native browser audio control for the selected track.
- the native control gives play, pause, volume, and timeline scrubbing without adding a custom waveform dependency yet.
- if the operator changes the focused track while the current focused track is playing, the newly focused track starts automatically.

- if the current focused track is paused, changing focus stays silent.

Consequences

- Local files remain the single source of truth, and the app can be relinked after new analysis/import work.
- The local database contains machine-specific absolute paths, so it is still ignored by git.
- Playback works through the same HTTP origin as the API and can later be replaced by object storage or managed media URLs without changing the client shape much.
- Native audio controls are enough for the current spike, but a future DJ workflow may add waveform previews, cue points, gain display, or keyboard shortcuts.

Further Considerations

- A persistent mini-player may be useful once focus changes become frequent, because it would separate "currently playing" from "currently inspected".
- The follow-playback behavior may eventually need an explicit UI toggle if the operator alternates between listening mode and catalog inspection mode.
- Waveform previews, cue markers, and beat-grid overlays should be considered only after the basic catalog/player flow feels stable.
- Server deployment will need a media-source strategy that replaces local absolute paths with managed storage paths or mounted media volumes.

Harmonic Transition Rules

DJ Desk treats every analyzed key as a transition profile. The profile is used for the draft-set Add warning and for compatibility coloring on the circle.

Profiles

Home

Regular major, natural minor, and raised-leading-tone minor tracks are home tracks. A major track is placed in the section of its relative minor.

```
C major -> La m section
A natural minor -> La m section
G natural minor with raised leading tone -> Sol m section
```

Pure Modal

Pure modal tracks keep their modal home tonic in the profile, but the UI places them in the section of the natural-minor pitch collection they use.

```
A Dorian pure:    home La m, displayed in Mi m collection
E Phrygian pure:  home Mi m, displayed in La m collection
C# Phrygian pure: home Do# m, displayed in Fa# m collection
```

Dorian and Lydian point clockwise from their home tonic to the collection section. Phrygian and Mixolydian point counter-clockwise. On the wheel this means the pure modal side zones around a boundary read as:

```
La natural minor
E pure Phrygian
La Dorian mixture + E Phrygian mixture
La pure Dorian
Mi natural minor
```

Modal Mixture

Modal-mixture tracks live on the boundary between their home section and modal target section. Neighboring modal-mixture tracks from both sides of the same boundary share one boundary zone.

```
A Dorian modal mixture:  La m / Mi m boundary
E Phrygian modal mixture: Mi m / La m boundary
C# Phrygian modal mixture: Do# m / Fa# m boundary
```

Directed Add Rules

The draft-set Add warning is directional: it checks whether the selected track can follow the last draft track.

- home -> home is valid when the sections are the same or adjacent.
- home -> pure modal is valid when the home section matches either the pure modal track's home section or target collection section.
- pure modal -> home is valid when the next home section matches either the pure modal track's home section or target collection section.
- pure modal -> pure modal is valid when both tracks use the same modal mode and their home sections are the same or adjacent.

- home -> modal mixture is valid when the home section belongs to that modal-mixture boundary.
- modal mixture -> home is valid when the home section belongs to that modal-mixture boundary.
- modal mixture -> modal mixture is valid only when both tracks use the same boundary.
- modal mixture -> pure modal is valid only into the same tonic and same mode.

No other cross-lane transition is considered valid in the current spike.

Reference Cases

D natural minor -> A natural minor	valid
D natural minor -> A Dorian pure	warning
D Dorian pure -> A Dorian pure	valid
A natural minor -> E natural minor	valid
A natural minor -> A Dorian pure	valid
A natural minor -> A Phrygian pure	valid
A natural minor -> E Dorian pure	warning
A natural minor -> E Phrygian pure	valid
A natural minor -> E Dorian mixture	warning
A natural minor -> A Dorian mixture	valid
A natural minor -> E Phrygian mixture	valid
A Dorian mixture -> E natural minor	valid
A Dorian mixture -> A natural minor	valid
A Dorian mixture -> A Dorian pure	valid
A Dorian mixture -> E Dorian pure	warning
A Dorian mixture -> E Phrygian mixture	valid
F# natural minor -> C# Phrygian mixture	valid
D natural minor -> A Dorian mixture	warning

Draft Reference Coloring

Circle coloring uses the last track in the draft set as the reference track, because the map should answer the planning question "what can I pick next?".

Selecting a row or a circle zone changes the focus panel and browser content, but does not change the compatibility colors.

A zone is compatible with the last draft track when at least one track in that zone can either follow the draft track or precede it. This symmetric rule keeps the map useful when browsing around the current draft endpoint.

Product Ideas

Focused Sector View

Add a second circle-display mode for detailed planning on small screens and dense key areas.

Instead of showing the whole circle, the view would show the active sector, its two neighboring sectors, and the two boundaries between them. This should make the areas that matter for the next transition much larger and easier to tap, especially on phones.

Open questions:

- Whether this mode should open on tap, via a segmented control, or both.
- Whether boundary modal-mixture zones should appear as full columns between sectors in this focused view.
- Whether the draft set should automatically move the focused view to the last added track's section.